

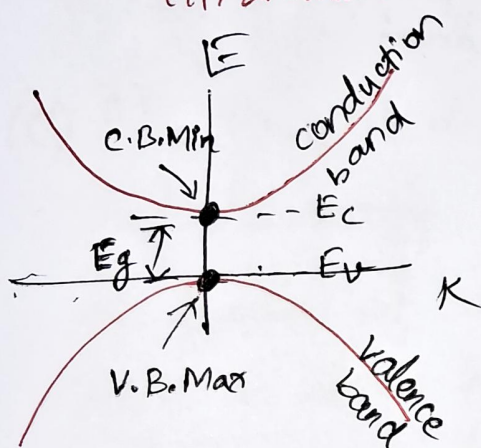
M2T1 Scheme of evaluation (PART-B)

③ Direct and indirect band gap semiconductors

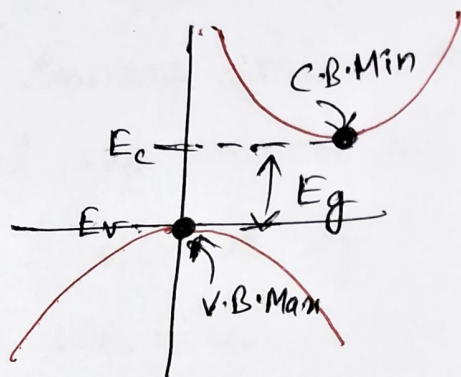
(a) Sketch the E vs k diagram for

(i) direct band gap semiconductor, and

(ii) indirect band gap semiconductor [2M]



Direct band gap s.c.
 E vs k diagram



Indirect band gap
s.c. E vs k diagram

C.B.Min = Conduction Band Minimum

V.B.Max = Valence Band Maximum

E_c = Energy level @ C.B.Min

E_v = Energy level @ V.B.Max

E_g = Energy band gap

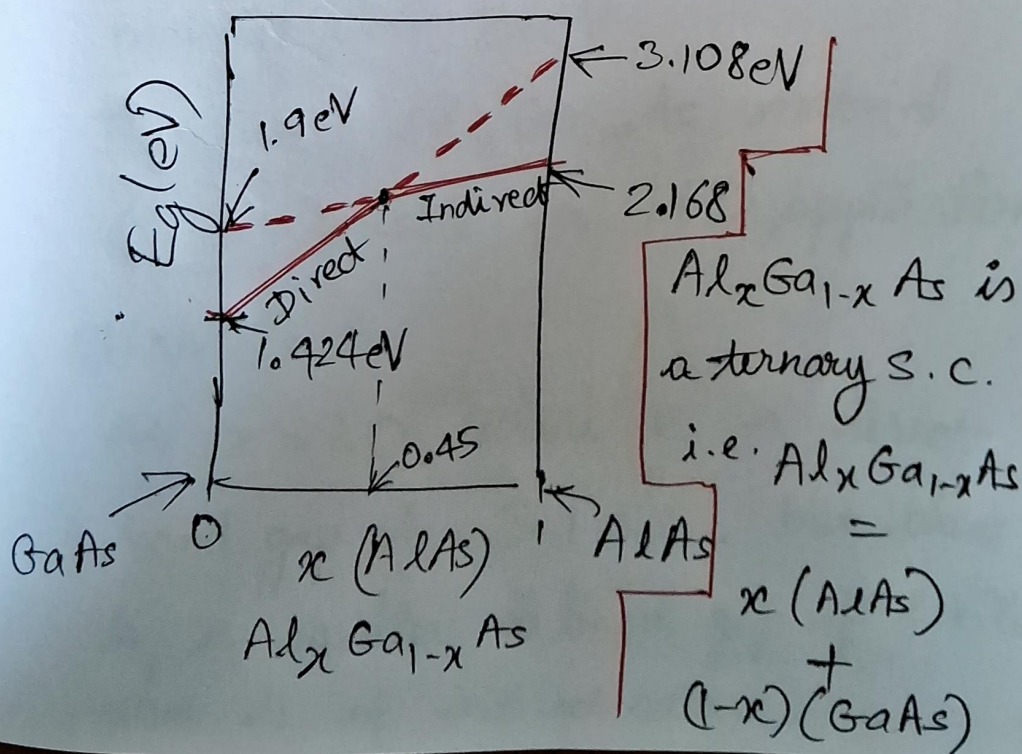
(b) Why is direct band gap s.c. useful for solar cell application? [2M]

Since direct band gap semiconductor does not require phonon for

absorption of photon, generation of electron-hole pair is an efficient process and we ^{have} high efficiency in conversion of light energy into electrical energy

photon + covalent bond $\rightarrow$ electron + hole

(C) $\text{Al}_x\text{Ga}_{1-x}\text{As}$ is a ternary semiconductor and the energy band gap variation as a function of mole fraction x of AlAs is shown in figure. Which composition can be used as material for blue LED application? [3M]



Blue photon has wavelength = 400nm
So energy of blue photon $= \frac{1.24 \text{ eV}}{\lambda(\mu\text{m})}$

$$E_{\text{blue}} = \frac{1.24 \text{ eV}}{0.4} = 3.1 \text{ eV}$$

Now $\text{Al}_x\text{Ga}_{1-x}\text{As}$ is a direct band gap semiconductor only from

$$x = 0 \quad \text{to} \quad x = 0.45$$

i.e. GaAs to $\text{Al}_{0.45}\text{Ga}_{0.55}\text{As}$

and bandgap varies from

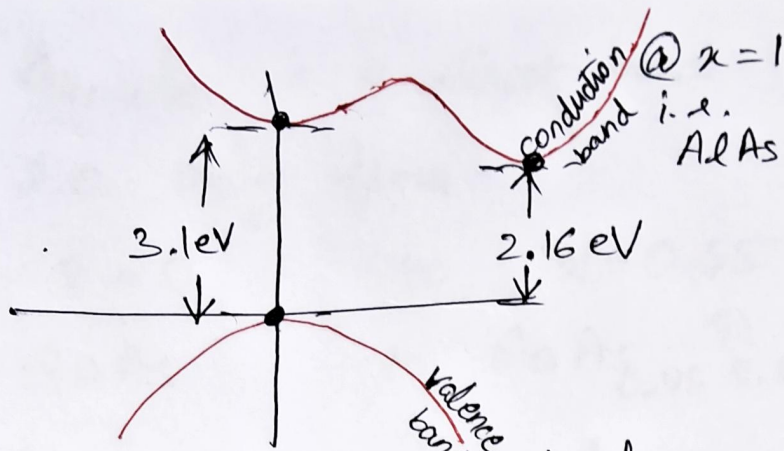
$$1.424 \text{ eV} \quad \text{to} \quad 1.9 \sim 2.0 \text{ eV}$$

But blue photon requires material with 3.1 eV.

Therefore, $\text{Al}_x\text{Ga}_{1-x}\text{As}$ material is not suitable for LED application

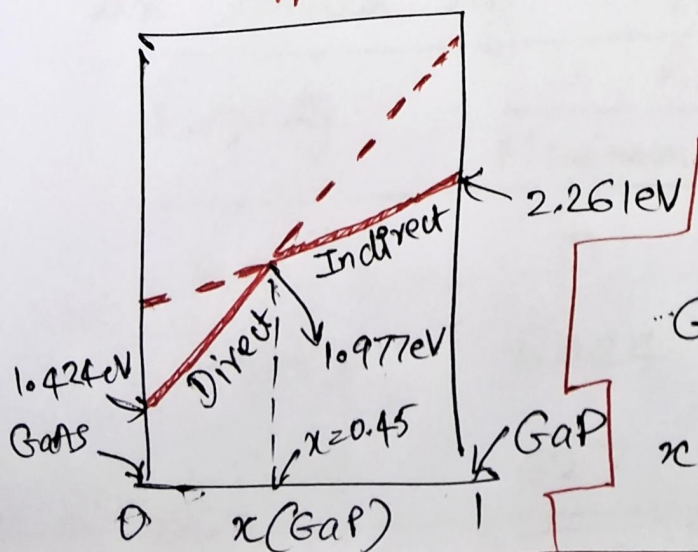
~~(A)~~ *NOTE*

At $x = 1.0$, there is a direct band gap of 3.1 eV, but there is a smaller ~~&~~ band gap of 2.168 eV that is of indirect nature.



E vs k diagram is of indirect type for $x=1$ i.e. AlAs material.

(d) The energy band gap variation of as a function of mole fraction x of GaP in the ternary semiconductor $\text{GaAs}_{1-x}\text{P}_x$ is shown in Figure below. What is the range of colours that can be emitted if this material is used for LED application? Report in wavelength. [3M]



$\text{GaAs}_{1-x}\text{P}_x$ is a ternary s.c. i.e.

$$\text{GaAs}_{1-x}\text{P}_x = x(\text{GaP}) + (1-x)(\text{GaAs})$$

GaAs_{1-x}P_x is a direct band gap s.c. only from

$$x=0 \quad \text{to} \quad x=0.45$$

i.e. GaAs to GaAs_{0.55}P_{0.45}

and band gap ~~varies~~ varies from

$$E_{\min} = 1.424 \text{ eV} \quad \text{to} \quad E_{\max} = 1.977 \text{ eV}$$

In terms of wavelength,

$$\lambda_{\max} = \frac{1.24}{1.424} = 0.871 \mu\text{m} \\ = 871 \text{ nm}$$

and

$$\lambda_{\min} = \frac{\cancel{1.24}}{\cancel{1.977}} \frac{1.24}{1.977} = 0.627 \mu\text{m} \\ = 627 \text{ nm}$$

Thus the range of colors that are emitted by GaAs_{1-x}P_x are

Property	Range	
	Minimum	Maximum
x	0	0.45
E (eV)	1.424	1.977
λ (nm)	627	871

— X — END OF PART-B